

STAT 17 Week 7

Central Limit Theorem (CLT) & Sampling Distributions

Xiao Zhang

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1. Sampling Distributions

Population parameters are fixed (but unknown). **Sample statistics** are random variables.

Population Parameter	Sample Statistic
Mean μ	$\bar{x}$
Std. Dev. σ	s
Proportion p	$\hat{p}$

Definition. A **sampling distribution** is the probability distribution of a statistic (e.g., $\bar{x}$ or $\hat{p}$) from all possible samples of size n .

2. CLT for the Sample Mean

Suppose:

- Simple random sample (SRS)
- Sample size n
- Population mean μ
- Population standard deviation σ

Center

$$\mu_{\bar{x}} = \mu$$

The sample mean is an unbiased estimator of μ .

Spread (Standard Error)

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

This is called the **standard error of the mean**.

As n increases:

Standard Error decreases

Sample means vary less than individual observations.

Shape

Case 1: Population is normal

$\bar{x}$ is normal for any n .

Case 2: Population not normal

Central Limit Theorem (CLT):

If $n \geq 30$ (general guideline),

$\bar{x} \approx \text{Normal}$

regardless of the population shape.

Larger $n \Rightarrow$ better approximation.

Full Model (When Conditions Met)

$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

3. Z-Score for Sample Means

If sampling distribution is approximately normal:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

Important distinction:

- Individual observation:

$$z = \frac{x - \mu}{\sigma}$$

- Sample mean:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

4. Conditions for CLT (Mean)

Always check:

1. Random sample

2. Independence:

$$n \leq 0.05N$$

3. Normality condition:

- Population normal OR
- $n \geq 30$

5. Sampling Distribution of the Sample Proportion

Suppose:

$$X \sim \text{Binomial}(n, p), \quad \hat{p} = \frac{X}{n}$$

Center

$$\mu_{\hat{p}} = p$$

Spread (Standard Error)

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

As n increases:

Standard Error decreases

Shape Condition

Sampling distribution of $\hat{p}$ is approximately normal if:

$$np(1-p) \geq 10$$

Full Model (When Conditions Met)

If:

- Random sample
- Independence
- $np(1-p) \geq 10$

Then:

$$\hat{p} \sim N\left(p, \sqrt{\frac{p(1-p)}{n}}\right)$$

6. Z-Score for Sample Proportions

If approximately normal:

$$z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

Note:

- Use population p in denominator.
- Check independence: $n \leq 0.05N$

7. Mean vs Proportion Summary

	Sample Mean	Sample Proportion
Center	μ	p
Spread	$\frac{\sigma}{\sqrt{n}}$	$\sqrt{\frac{p(1-p)}{n}}$
Normal Condition	$n \geq 30$	$np(1-p) \geq 10$
Z Formula	$\frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$	$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$